Heriot-Watt University

Department of Actuarial Mathematics and Statistics

Life Insurance Mathematics B

Tutorial 2

1. (Proving recursive relationships between policy values) State and prove the recursive relationship between policy values of (i) a pure endowment with level annual premiums; (ii) a level annuity payable annually in advance for life.

2. (Excel exercise on recursive policy values) To complete this exercise, start from the file *t2q2 (Recursive policy value calcs) - start.xlsx* on Canvas. The tables you need are included in the starting spreadsheet.
   (a) A life office sells non-profit endowment assurances with annual premiums and term 25 years to lives age 30. It calculates net premium policy values assuming the mortality of the A1967-70 ultimate table and 4% p.a. interest. Calculate the policy values at all integer durations using the recursive relationship starting with $V(25) = 1$.
   (b) A life office sells term assurances with annual premiums and term 25 years to lives age 30. It calculates net premium policy values assuming the mortality of the A1967-70 ultimate table 4% p.a. interest. Calculate the policy values at all integer durations using the recursive relationship starting with $V(25) = 0$.

3. (Understanding and interpreting policy values) It is possible for a policy value to be negative. For example, a person with a mortgage or other loan might take out a life insurance policy with a decreasing sum assured so that if they die, the sum assured is sufficient (at least approximately) to pay off the loan. The purpose might be to protect the value of the estate they pass on to dependants or it might be a condition of the loan that they must take out life cover. If the premium is level but the sum assured is decreasing, that can lead to a negative policy value. What does it mean for a policy value to be negative? Why might it be a problem for an insurer? And what could the insurer do about it?

4. (Calculating policy values recursively) Calculate the following on the basis of AM92 Ultimate mortality and 4% interest, stating any approximations you use: For a 25-year endowment sold to a man age 35, with sum assured £1 paid at the end of the year of death or on maturity, and with level quarterly premiums: $V(16.75)$ and $V(16.5)$.

5. (Thiele's DE proofs in special cases) State and prove Thiele's differential equation for the following contracts, and in each case give the appropriate boundary condition at maturity/expiry.
   (a) A temporary annuity with term n years, payable continuously at rate £1 per annum to a life now age x as long as they are alive during the term.
   (b) A pure endowment with term n years, and sum assured £1 payable to a life now age x if they survive to the end of the term. There is a single premium paid at outset.
   (c) A deferred annuity sold to a life now age x, paid for by a single premium. Payment is continuous at rate £1 per annum starting in n years, if the person is then alive, and then continuing for life.

6. (Excel exercise using Thiele's DE and an Euler scheme): You may find it helpful to start from the file *t2q6 (Euler and Thiele) - start.xlsx*.

   Suppose the force of mortality is given by the Gompertz function:

$$\mu_x = 0.0004 \times 1.09^x.$$

   (a) Last semester, you were given an exact formula for ${}_tp_x$ derived from the Gompertz function. But instead, using an Euler scheme with step size $h = 0.01$ years to solve the Kolmogorov equation, find approximate values of ${}_tp_{20}$ for $0 \leq t \leq 10$. How does the approximate value of ${}_{10}p_{20}$ compare with the correct value?
   (b) A person age 20, subject to the force of mortality above, buys a 10-year temporary assurance contract, with sum assured £1 payable immediately on death. The premium rate is payable continuously at rate £0.003 per annum. The force of interest is 0.05 per annum. Using an Euler scheme with step size $h = 0.01$ years to solve Thiele's differential equation, show that the policy value at outset, $V(0)$ is approximately £0.0033.
   (c) Explain why, in this case, $V(0) \neq 0$.
   (d) By trying different values for the annual rate of premium, find the rate of premium that results in $V(0) = 0$.

7. (Quick question writing down Thiele DE) A man age x buys a special endowment assurance with term n years, with the following benefits:

   - On maturity, a sum assured of £10,000.
   - On death before maturity, a sum assured of £10,000 *plus* the policy value at the time of death.

   Premiums are payable continuously at rate P per annum. Write down Thiele's differential equation for this policy, and state what boundary condition you would use in its solution.

8. (Excel exercise using Euler schemes, and Thiele and Kolmogorov DEs): The Excel file *t2q7 (Euler Thiele and Kolm)- start.xlsx* contains values of μ_x for the AM92 ultimate and AF92 ultimate life tables, at intervals of 1 year, 0.1 year and 0.01 year. (The last two have been found approximately by linear interpolation between the values at integer ages.)

   (a) Using Euler schemes with step sizes 1 year, 0.1 year and 0.01 year to solve the Kolmogorov equation, find approximate values of ${}_5p_{30}$, ${}_{10}p_{30}$ and ${}_{15}p_{30}$ for males only. How accurate are these, compared with the correct values found from the Yellow Tables?

   (b) Suppose a man age 30 buys the endowment assurance described in Q7, with term 15 years. The life office uses the same basis for calculating premiums and policy values, namely a force of interest of 0.04 per annum and the AM92 ultimate life table. Find the rate of premium for which $V(0) = 0$ by solving Thiele's differential equation with step size 1 year, 0.1 year and 0.01 year.

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F70LB Tutorial 2

Optional questions

The questions in this section are all optional. They have been retained from older versions of the course to give students additional practice if they wish. Solutions will be provided, but it is not planned to cover them in tutorials. Your priority should be to cover all the main written and Excel questions before attempting any of these.

1. A life insurer issued a special endowment insurance policy to a life age 40, with term 5 years. The death benefit, payable at the end of the year of death, is equal to £1,000 *plus* the reserve that would have been held at the end of the year had the policyholder been alive. The maturity benefit is £1,000. The premium is £179.30, payable annually in advance. By using the recursive relation between reserves, or otherwise, find the reserve just before payment of the 3rd premium. The basis for all calculations is A1967–70 Ultimate mortality and 4% interest.

2. A man age x buys a special endowment assurance with term n years, with the following benefits:

   - On maturity, a sum assured of £10,000.
   - On death before maturity, a sum assured of £10,000 *plus* the policy value at the time of death.

   Premiums are payable continuously at rate P per annum. Write down Thiele's differential equation for this policy, and state what boundary condition you would use in its solution.